H Dharshan

dharshan080622@gmail.com +91 70693 18874 Dharshan2208 Dharshan2208 Portfolio

Summary

Computer Science undergraduate experienced in backend systems, distributed architectures, and AI applications. Built scalable services, REST APIs, and multi agent workflows using Go, Python, FastAPI, React, Redis, PostgreSQL, and Docker. Strong foundation in data structures and algorithms with competitive programming experience as a Codeforces Pupil (1212 rating).

Education

Amrita Vishwa Vidyapeetham, Coimbatore *Aug 2024 – Aug 2028*
B.Tech in Computer Science **CGPA: 8.64**

Projects

Judex: *Go, Redis, Docker, REST APIs, Azure (Jun 2026)* GitHub Repository

- Designed a **distributed online code execution service** that supports secure multi language compilation and execution through Docker based sandboxing with resource isolation, execution time limits, and RESTful APIs.
- Architected an **asynchronous job processing pipeline** using Redis backed queues and worker services, implemented execution scheduling, job tracking, and **IP based rate limiting** to enable scalable, reliable, and abuse resistant code execution.
- Integrated Prometheus, Grafana, Loki, and Promtail for end-to-end observability, validated platform scalability through **k6** load testing with **1,000 concurrent users and 50K+ total requests without service failures**, and deployed the distributed system on **Microsoft Azure** using Docker.

Git Secret Scanner: *Go, Git, Concurrency, Security, CLI (Jul 2026)* GitHub Repository

- Developed a **concurrent Git security scanner** in Go that analyzes both **source code and full commit history** to detect exposed credentials using a **multi layer pipeline** combining regex signatures, entropy analysis, and heuristic keyword detection.
- Engineered a **worker pool based concurrent architecture** enabling scalable repository scanning with full history traversal and optimized resource utilization.
- Achieved high-performance execution, scanning **100K+ LOC repositories in 8 seconds** and medium repositories with 50 commits in 15 seconds, while generating deduplicated security reports.

Stock Analysis System: *Python, LangGraph, Gemini, Yahoo Finance (Nov 2025)* GitHub Repository

- Engineered a **multi agent investment analysis system** using LangGraph where specialized AI agents collaboratively evaluated financial data through **structured reasoning** and **consensus-based decision making**.
- Integrated Gemini, Groq, Yahoo Finance, and financial APIs to retrieve real time market data, financial statements, company fundamentals, and news for **evidence based investment analysis**.
- Designed a **structured reasoning pipeline** that synthesized multi agent outputs into comprehensive investment reports containing risk assessment, valuation insights, and actionable recommendations.

Vita Care: *React.js, FastAPI, Gemini AI, WhatsApp Integration (Jan 2026)* GitHub Repository

- Developed an **AI assisted healthcare coordination platform** leveraging React, FastAPI, and Gemini AI to automate appointment scheduling, prescription management, and medication reminders.
- Designed a modular microservices architecture integrating **autonomous AI workflows**, voice interaction, and **WhatsApp messaging** to streamline patient communication.

Technical Skills

Programming Languages: Go, Python, Java, JavaScript, C, C++
Frameworks & Libraries: React.js, FastAPI, LangChain, LangGraph
Databases: MySQL, PostgreSQL, Redis, Supabase
Tools & Technologies: Docker, Git, Github, Linux, CI/CD, Cloud Deployment(Azure)

Achievements

- **Codeforces Pupil** with a peak rating of **1212**.
- Solved **300+ coding problems** across **Codeforces**, **LeetCode**, and **CodeChef**.